liquidity providers (LPs) earn yield by depositing assets into a pool because the pool takes a fee for every trade (LPs benefit from high trading volume). This allows the pool to attract liquidity but exposes LPs to smart contract risk and impermanent loss, which occurs when two assets in a pool have uncorrelated returns and high volatilities.[19] These properties allow arbitrageurs to profit from the asset volatilities and price differences, reducing the temporary returns for LPs and exposing them to risk if an asset moves sharply in price. Some AMMs, such as Cap,[20] are able to reduce impermanent loss by using an oracle to determine exchange prices and dynamically adjusting a pricing curve to prevent arbitrageurs from exploiting LPs, but impermanent loss remains a large problem with most AMMs used today.

On May 5, 2021, Uniswap lauched its third version. The key difference between v2 and v3 is that liquidity providers can allocate funds to a custom range (the range in the CFMM is not limited and potentially infinite). This creates individualized price curves and traders interact with the aggregation of the liquidity of all of these curves. Given the ability to specify a range, v3 is somewhat analogous to a limit order system.

On-chain order-book DEXs have a different but prevalent set of risks. These exchanges suffer from the scalability issues inherited from the underlying blockchain they run atop of and are often vulnerable to front-running by sophisticated arbitrage bots. Order-book DEXs also often have large spreads due to the presence of low-sophistication market makers. Whereas traditional finance is able to rely